```cpp
// Task 1: Array Traversal
include <iostream>
using namespace std;
int main() {
    int n;
    int arr[100];
    int sum = 0;
    cout << "Enter number of elements: ";
    cin >> n;
    cout << "Enter " << n << " elements:\n";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
        sum += arr[i];
    }
    cout << "\nArray Elements and Indices:\n";
    for (int i = 0; i < n; i++) {
        cout << "Index " << i << " = " << arr[i] << endl;
    }
    double average = (double)sum / n;
    cout << "\nSum = " << sum << endl;
    cout << "Average = " << average << endl;

    return 0;
}
```

```cpp
// Task 2: Array Insertion
#include <iostream>
using namespace std;
int main() {
    int arr[100], n, element, index, choice;
    cout << "Enter number of elements: ";
    cin >> n;
    cout << "Enter " << n << " elements:\n";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }
    cout << "\nArray before insertion: ";
    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }
    cout << "\n\nWhere do you want to insert?\n";
    cout << "1. Beginning\n";
    cout << "2. End\n";
    cout << "3. Given Index\n";
    cout << "Enter your choice: ";
    cin >> choice;
    cout << "Enter element to insert: ";
    cin >> element;
    if (choice == 1) {
        for (int i = n; i > 0; i--) {
            arr[i] = arr[i - 1];
        }
        arr[0] = element;
        n++;
    }
    else if (choice == 2) {
        arr[n] = element;
        n++;
    }
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    else if (choice == 3) {
        cout << "Enter index (0 to " << n << "): ";
        cin >> index;
        if (index < 0 || index > n) {
            cout << "Invalid index!";
            return 0;
        }
        for (int i = n; i > index; i--) {
            arr[i] = arr[i - 1];
        }
        arr[index] = element;
        n++;
    }
    else {
        cout << "Invalid choice!";
        return 0;
    }
    cout << "\nArray after insertion: ";
    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;
    return 0;
}
```

```
//Task 3: Array Deletion
#include <iostream>
using namespace std;
int main() {
    int arr[100], n, index, choice;
    cout << "Enter number of elements: ";
    cin >> n;
    cout << "Enter " << n << " elements:\n";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }
    cout << "\nArray before deletion: ";
    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }
    cout << "\n\nWhere do you want to delete?\n";
    cout << "1. Beginning\n";
    cout << "2. End\n";
    cout << "3. Given Index\n";
    cout << "Enter your choice: ";
    cin >> choice;

    if (n == 0) {
        cout << "Array is empty!";
        return 0;
    }
    if (choice == 1) {
        for (int i = 0; i < n - 1; i++) {
            arr[i] = arr[i + 1];
        }
        n--;
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    }
    else if (choice == 2) {
        n--;
    }
    else if (choice == 3) {
        cout << "Enter index (0 to " << n - 1 << "): ";
        cin >> index;
        if (index < 0 || index >= n) {
            cout << "Invalid index!";
            return 0;
        }
        for (int i = index; i < n - 1; i++) {
            arr[i] = arr[i + 1];
        }
        n--;
    }
    else {
        cout << "Invalid choice!";
        return 0;
    }
    cout << "\nArray after deletion: ";
    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;
    return 0;
}
```

```cpp
//Task 4: Searching and Sorting
#include <iostream>
using namespace std;
int main() {
    int arr[100], n, search, position = -1;
    cout << "Enter number of elements: ";
    cin >> n;
    cout << "Enter " << n << " elements:\n";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }
    cout << "\nOriginal Array: ";
    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }
    cout << "\n\nEnter element to search: ";
    cin >> search;
    for (int i = 0; i < n; i++) {
        if (arr[i] == search) {
            position = i;
            break;
        }
    }
    if (position != -1) {
        cout << "Element found at index: " << position << endl;
    } else {
        cout << "Element not found!" << endl;
    }
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    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
    cout << "\nArray after Bubble Sort (Ascending): ";
    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;
    return 0;
}
```

```cpp
// Task 5: Menu-Driven Array Operations
#include <iostream>
using namespace std;
int main() {
    int arr[100], n, choice, element, index, key;
    bool found;
    cout << "Enter number of elements: ";
    cin >> n;
    if (n < 0 || n > 100) {
        cout << "Invalid number of elements!";
        return 0;
    }
    cout << "Enter " << n << " elements:\n";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }
    do {
        cout << "\n\n===== ARRAY MENU =====\n";
        cout << "1. Display Array\n";
        cout << "2. Insert Element\n";
        cout << "3. Delete Element\n";
        cout << "4. Search Element\n";
        cout << "5. Sort Array\n";
        cout << "6. Exit\n";
        cout << "Enter your choice: ";
        cin >> choice;
        switch (choice) {
        case 1:
            if (n == 0) {
                cout << "Array is empty!";
            } else {
                cout << "Array: ";
                for (int i = 0; i < n; i++) {
                    cout << arr[i] << " ";
                }
            }
            break;
        case 2:
            if (n == 100) {
                cout << "Array is full! Cannot insert.";
                break;
            }
            cout << "Enter element: ";
            cin >> element;
            cout << "Enter index (0 to " << n << "): ";
            cin >> index;
            if (index < 0 || index > n) {
                cout << "Invalid index!";
            } else {
                for (int i = n; i > index; i--) {
                    arr[i] = arr[i - 1];
                }
                arr[index] = element;
                n++;
                cout << "Element inserted successfully!";
            }
            break;
        case 3:
            if (n == 0) {
                cout << "Array is empty! Cannot delete.";
                break;
            }
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            cout << "Enter index to delete (0 to " << n - 1 << "): ";
            cin >> index;
            if (index < 0 || index >= n) {
                cout << "Invalid index!";
            } else {
                for (int i = index; i < n - 1; i++) {
                    arr[i] = arr[i + 1];
                }
                n--;
                cout << "Element deleted successfully!";
            }
            break;
        case 4:
            if (n == 0) {
                cout << "Array is empty!";
                break;
            }
            cout << "Enter element to search: ";
            cin >> key;
            found = false;
            for (int i = 0; i < n; i++) {
                if (arr[i] == key) {
                    cout << "Element found at index " << i << ".";
                    found = true;
                    break;
                }
            }
            if (!found) {
                cout << "Element not found!";
            }
            break;
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        case 5:
            if (n == 0) {
                cout << "Array is empty!";
                break;
            }
            for (int i = 0; i < n - 1; i++) {
                for (int j = 0; j < n - i - 1; j++) {
                    if (arr[j] > arr[j + 1]) {
                        int temp = arr[j];
                        arr[j] = arr[j + 1];
                        arr[j + 1] = temp;
                    }
                }
            }
            cout << "Array sorted in ascending order!";
            break;
        case 6:
            cout << "Program exited successfully.";
            break;
        default:
            cout << "Invalid choice! Please enter a number between 1 and 6.";
        }
    } while (choice != 6);
    return 0;
}

```